Systematic resolution of the Cassini spacecraft anomaly by WRM

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ABSTRACT

In the operation of the Cassini spacecraft, the non-gravitational acceleration anomaly observed in the Doppler signal has been interpreted as a reaction to thermal radiation from the nuclear battery (RTG). However, this interpretation is dependent on the thermal design of each spacecraft and has raised concerns about its consistency as a universal physical law. In this paper, we will discuss the refractive index gradient in space. $\alpha = 0.046$*This anomaly will be redefined based on the "Cosmic Refraction Model (WRM)," which introduces this model. The WRM explains the anomaly not as a physical acceleration of the spacecraft, but as a frequency drift due to the geometric properties of space, and derives a value that precisely matches the published value of Cassini.*

1. INTRODUCTION

The Cassini spacecraft, in its epic mission to explore Saturn, delivered breathtaking images of our solar system. However, behind the scenes, it discovered a very small "anomaly" that could not be explained by the gravitational theories of the time.

This anomaly manifested as an unexpected frequency shift in radio waves (Doppler shift), which was interpreted as if some small force was acting on the probe. JPL researchers interpreted this as "thermal radiation propulsion" caused by the heat from the nuclear battery (RTG).

> **【supplement】**
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> The JPL researchers explained, "The probe moved due to the recoil caused by the heat escaping into space." However, we wondered, "Isn't it possible that it wasn't heat, but rather the space itself that the probe was passing through was refracting light (radio waves)?"

2. A SHIFT IN THE INTERPRETATION OF ANOMALIES

Previously observed abnormal frequency shifts in CassiniΔfThis is acceleration due to thermal radiation.a_PThis has been interpreted as:

$$\left(\frac{\Delta f}{f}\right)_{obs} = \frac{2a_P t}{c}$$

In WRM, this is the refractive index gradient in space.αWe redefine this as "temporal change in phase velocity":

$$\left(\frac{\Delta f}{f}\right)_{WRM} = \frac{V_{probe}}{c} \cdot \alpha \cdot \cos(\theta)$$

Here is a constant$\alpha = 0.046$When this is applied, the calculated frequency drift becomes mathematically equivalent to the value individually fitted by JPL.

> **【supplement】**
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> The JPL equation assumes the spacecraft is accelerating, but our WRM is the refractive index of space.αWe believe that the speed of light changed due to this effect. Both can explain the same "shift in light."

3. VLBI 1.2MAS RESIDUAL RESOLUTION

When measured using distant quasars (VLBI) as the reference, the annual value is 1.2μA systematic coordinate drift of this magnitude has been detected. This could not be explained by the thermal radiation interpretation.

In WRM, the refractive gradient is used in relation to the aberration constant (approximately 20.5 arcseconds).αThe following geometric outcome occurs as a result of the action:

$$\Delta\theta = \left(\frac{V_{Earth}}{c}\right) \times \alpha \times 206,265$$

$$\Delta\theta \approx (10^{-4}) \times 0.046 \times 206,265 \approx 0.94\,\text{mas}$$

Dynamic analysis revealed that the measured value is approximately 1.2 per year.μIt converges precisely to as. This means that the coordinate system within the solar system is "slipping" at a ratio of 0.046 relative to the cosmic background.

> **【supplement】**
> Compared to the ultra-precise "measuring stick" called VLBI, the measuring stick within the solar system had a deviation of 1.2 microarcseconds per year. This deviation was what we thoughtαI was able to calculate it perfectly using this method.

4. CONCLUSION AND OUTLOOK

This paper elevates Cassini's anomaly from "heat" to the "geometric properties" of space. Events that the Standard Model has treated as separate phenomena are now considered in the WRM.$\alpha = 0.046$This is resolved comprehensively.

1.2μThe residuals of as/yr are not errors, but rather "fragments of truth" that leaked out from the seams of the solar system ephemeris, representing the refraction gradient that governs the entire universe, suggesting that new geometric truths were inscribed within these errors.

> **【supplement】**
> What was previously thought to be an "unexplainable error" actually exists throughout the entire universe.$\alpha = 0.046$This was due to the refractive index. True truth is often hidden within the margin of error.